

Vakuum enačen v vakuumu
- produkti derivacij $\vec{E}$ in $\vec{B}$ v prostoru ni izvirni ($\vec{j} = 0$)

$$\begin{aligned} \nabla \cdot \vec{E} &= 0 & \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} & \nabla \times \vec{B} &= \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

$$\downarrow \nabla \times$$

$$\nabla \times (\nabla \times \vec{E}) = \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\nabla^2 \vec{E} = -\frac{\partial}{\partial t} \nabla \times \vec{B} \rightarrow \nabla^2 \vec{E} - \epsilon_0 \mu_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

$$\nabla^2 \vec{B} - \epsilon_0 \mu_0 \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

hitrost svetlobe: $c^2 = \frac{1}{\epsilon_0 \mu_0}$

- rešitev val. en. (ravni valovi)

$$\vec{E}(\vec{r}, t) = \vec{E}(\vec{r}) e^{i\omega t} \rightarrow \omega = kc$$

$$\vec{B}(\vec{r}, t) = \vec{B}(\vec{r}) e^{i\omega t}$$

$$\rightarrow \vec{E}(\vec{r}, t) = \text{Re} \left[\sum_n \vec{E}_n e^{i(\vec{k} \cdot \vec{r} - \omega_n t)} \right]$$

↳ lahko kompleksno

$$\vec{B}(\vec{r}, t) = \text{Re} \left[\sum_n \vec{B}_n e^{i(\vec{k} \cdot \vec{r} - \omega_n t)} \right]$$

→ ker prostor brez izvirov

$$0 = \nabla \cdot \vec{E} = \sum_n i \vec{k} \cdot \vec{E}_n e^{i(\vec{k} \cdot \vec{r} - \omega_n t)}$$

$$\Rightarrow \vec{k} \cdot \vec{E}_n = 0 \Rightarrow \boxed{\vec{k} \perp \vec{E}_n}$$

$$0 = \nabla \cdot \vec{B} = \sum_n i \vec{k} \cdot \vec{B}_n e^{i(\vec{k} \cdot \vec{r} - \omega_n t)}$$

$$\Rightarrow \vec{k} \cdot \vec{B}_n = 0 \Rightarrow \boxed{\vec{k} \perp \vec{B}_n}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = \sum_n i\omega \vec{B}_n e^{i(\vec{k} \cdot \vec{r} - \omega_n t)}$$

$$\sum_n i \vec{k} \times \vec{E}_n e^{i(\vec{k} \cdot \vec{r} - \omega_n t)}$$

$$\Rightarrow \vec{k} \times \vec{E}_n = \omega \vec{B}_n \Rightarrow \boxed{\vec{E}_n \perp \vec{B}_n}$$